

Predicting Alzheimer's Risk Using Protein Biomarkers, Demographics, and TBI History Through Machine Learning

Bitanya Darge (Dr. Stough, advisor)



Introduction

Alzheimer's disease (AD) develops slowly, often years before memory decline, and early detection remains a major challenge in neuroscience and clinical care. Biological markers such as tau, phosphorylated-tau (p-tau), amyloid-beta (A β 42/A β 40), inflammation proteins, and demographic risk factors (age, APOE genotype, education) are known to play key roles in AD progression. Additionally, traumatic brain injury (TBI) has been associated with long-term neurodegeneration and elevated AD risk.

Machine learning provides a powerful approach for integrating multiple biomarkers and demographic factors at once, capturing subtle nonlinear patterns that traditional statistical methods often miss. Early screening models may help identify individuals at elevated risk before diagnosis is possible.

The goal of this project was to explore whether machine learning models can accurately distinguish individuals with Alzheimer's Disease from those without dementia using:

- **Protein biomarkers** (tau, p-tau, amyloid, cytokines)
- **Demographics** (age, sex, education, APOE4 status)
- **TBI history** (loss of consciousness, number of TBIs, age at first injury)

This project evaluates and compares two models **Logistic Regression** and **Random Forest** to determine which provides better predictive performance and which biological factors most strongly contribute to Alzheimer's risk.

Methods

A dataset of **345 individuals** from the Allen Brain Atlas dataset was analyzed. Diagnostic labels were simplified into: **No Dementia (0)** vs. **Possible/Probable Alzheimer's Disease (1)**.

1. **Data preprocessing:** Missing values imputed; numeric features standardized; categorical variables one-hot encoded.
2. **Train/test split:** 80/20 stratified split for evaluation.
3. **Models:** Logistic Regression (linear baseline) and Random Forest (400-tree nonlinear model).
4. **Validation:** Accuracy, ROC-AUC, PR-AUC, precision, recall, and F1-score.
5. **Feature importance:** Random Forest used to identify key biological and demographic predictors.

Results

Two machine learning models were evaluated to compare linear versus nonlinear predictive performance.

The **Logistic Regression** model performed slightly above chance, achieving an accuracy of **0.58**, ROC-AUC of **0.585**, and PR-AUC of **0.580**. While precision and recall were balanced across classes, the model struggled to capture the nonlinear interactions present among biomarkers and risk factors.

In contrast, the **Random Forest** model demonstrated substantially stronger performance. It achieved an accuracy of **0.75**, ROC-AUC of **0.840**, and PR-AUC of **0.759**, showing much better discrimination between Alzheimer's and non-demented individuals. Feature importance analysis revealed that age, education level, tau-related proteins, amyloid markers (A β 40, A β 42), inflammatory cytokines (e.g., IL-6, MCP-1, RANTES), and TBI history all contributed meaningfully to prediction, consistent with established biological literature.

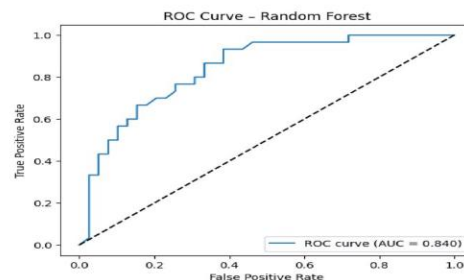


Figure 1. ROC Curve – Random Forest

Higher AUC indicates stronger ability to distinguish Alzheimer's vs. non-demented individuals.

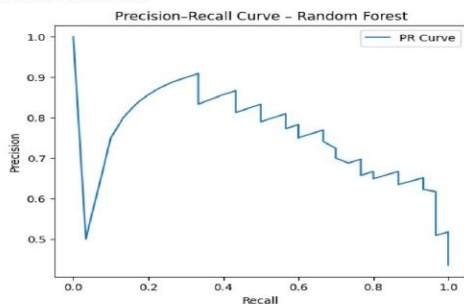


Figure 2. Precision–Recall Curve – Random Forest

A valuable metric for imbalanced datasets, showing strong precision maintained across varying recall levels.

Discussion

The superior performance of the Random Forest model suggests that Alzheimer's risk depends on complex, nonlinear relationships between biological and demographic variables. Tau and amyloid biomarkers, along with inflammatory cytokines, emerged as important predictors, supporting existing literature linking protein pathology and neuroinflammation to AD progression. The importance of education supports the "cognitive reserve" theory, while TBI-related predictors reinforce research suggesting long-term neurodegenerative consequences of head injuries.

These findings highlight the potential of integrating machine learning with proteomic and demographic data to improve early risk screening. However, the use of cross-sectional postmortem data limits the ability to predict disease progression over time. The dataset's modest size and class imbalance may also influence performance. Future studies should incorporate longitudinal follow-up, additional biomarkers (e.g., NfL), and larger sample sizes to further validate predictive accuracy and clinical applicability.

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